

Literature Review: Variance Risk Premium in the Cross-Section of Returns

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1 Introduction

Our previous empirical study on the China 50ETF documented two striking findings. First, using the Bollerslev, Tauchen, and Zhou (2009, hereafter BTZ) regime classification, VRP has near-zero predictive power in low- and medium-volatility states but achieves an R^2 of 6.14% in high-volatility regimes. Second, among 20 VRP-derived factors, the rolling Z-score (`vrp_zscore`) dramatically outperforms the raw VRP level (6.14% vs. 2.15% R^2), followed by nonlinear terms such as `vrp_cubic` ($R^2 = 4.21\%$) and `vrp_squared` ($R^2 = 3.95\%$).

These findings raise three questions that the literature can help answer:

1. Is VRP a priced factor in the cross-section of individual stock returns, beyond the standard market, size, and value factors?
2. Why does the nonlinear (Z-score) transformation outperform the raw VRP?
3. How should a Fama–MacBeth cross-sectional regression be designed to test VRP’s incremental pricing power?

2 Han and Zhou (2011): VRP and the Cross-Section of Stock Returns

2.1 Research Design

Han and Zhou (2011) provide the first in-depth cross-sectional analysis of individual stock VRP. They define the VRP for each stock i in month t as:

$$VRP_{i,t} = IV_{i,t} - RV_{i,t} \tag{1}$$

where $IV_{i,t}$ is the risk-neutral expected variance implied by equity option prices (constructed using the model-free methodology of Britten-Jones and Neuberger, 2000), and $RV_{i,t}$ is the realized variance estimated from high-frequency (5-minute) stock returns.

Their empirical strategy proceeds in two stages:

1. **Portfolio sorts:** At the end of each month, all stocks with available option data are sorted into decile portfolios based on their VRP. Value-weighted and equal-weighted portfolio returns are computed for the subsequent month.
2. **Fama–MacBeth regressions:** Monthly cross-sectional regressions of individual stock returns on VRP and control variables, including market beta, log market capitalization (size), book-to-market ratio, past returns (momentum), and idiosyncratic volatility.

2.2 Key Findings

The high-minus-low VRP decile spread produces an average monthly return of 2.00% (value-weighted) and 1.75% (equal-weighted). This spread is robust to controlling for the Fama–French three factors, the Carhart momentum factor, and various firm characteristics. In Fama–MacBeth regressions, VRP retains a positive and statistically significant coefficient after controlling for market beta, size, and book-to-market ratio, confirming that VRP is an independently priced factor.

2.3 Relevance to Our Study

Han and Zhou’s framework is the most direct precedent for our planned Fama–MacBeth test. However, two important gaps exist. First, they use *raw* VRP as the cross-sectional variable. Our prior results suggest that nonlinear transformations (particularly the Z-score) may yield substantially stronger cross-sectional pricing power. Second, their analysis is conducted on U.S. equities with a relatively small sample. Extending this framework to the Chinese market, where option markets are younger and the cross-section of optionable stocks is narrower, is a nontrivial contribution.

3 Eraker and Wang (2015): Nonlinear Dynamics of VRP

3.1 The Nonlinear Diffusion Model

Eraker and Wang (2015) propose a class of nonlinear diffusion processes obtained as transformations of basic affine processes. In their framework, the VIX index (or its square) is modeled as a nonlinear diffusion $Y_t = g(X_t)$, where X_t follows a standard affine (e.g., square-root) process and $g(\cdot)$ is a monotone nonlinear transformation. The key innovation is a *cubic drift term*: while standard affine models imply linear mean reversion, the nonlinear model allows the speed of mean reversion to vary with the level of Y_t .

The VRP is defined as the difference between the drift under the risk-neutral measure $\mathbb{Q}$ and the physical measure $\mathbb{P}$:

$$VRP_t = \mu^{\mathbb{Q}}(Y_t) - \mu^{\mathbb{P}}(Y_t) \quad (2)$$

This definition naturally produces a state-dependent VRP: because the drifts are nonlinear functions of Y_t , the VRP varies nonlinearly with the current volatility level.

3.2 Key Findings

Using maximum likelihood estimation on VIX index levels (under $\mathbb{P}$) and VIX futures prices (under $\mathbb{Q}$), Eraker and Wang find:

1. **Significant nonlinearity under both measures:** The data strongly reject linear (affine) specifications in favor of the nonlinear model.
2. **Asymmetric mean reversion:** The VIX reverts more quickly from high levels than from low levels. This implies that crisis regimes are *less persistent* than a linear model would predict.
3. **Stronger predictive power:** The nonlinear VRP measure has strong out-of-sample predictive power for aggregate stock returns, outperforming the standard linear VRP of BTZ (2009).

3.3 Relevance to Our Study

Eraker and Wang’s findings provide the theoretical foundation for two key results in our prior report:

Why VRP only works in high-volatility regimes. The nonlinear mean reversion implies that the gap between $\mathbb{Q}$ and $\mathbb{P}$ expectations—i.e., the VRP—is mechanically amplified in high-volatility states. In low-volatility states, the nonlinear drift is approximately linear, and the $\mathbb{P}$ – $\mathbb{Q}$ wedge is small and dominated by noise. This provides a formal mechanism for our BTZ regime result beyond the heuristic “insurance premium” explanation.

Why the Z-score outperforms raw VRP. Raw VRP is an absolute measure in variance units. In a nonlinear model, a VRP of 100 in a calm market conveys fundamentally different information than a VRP of 100 during a crisis, because the baseline level, mean-reversion speed, and volatility-of-volatility are all state-dependent. The Z-score standardization ($Z_t = (VRP_t - V\bar{R}P_{t-60:t})/\sigma_{VRP,t-60:t}$) implicitly conditions on the recent volatility regime, effectively extracting the “surprise” component of VRP. This is analogous to isolating the *jump risk* component emphasized by Eraker and Wang: sudden, unexpected deviations from the locally adapted volatility level. Importantly, the strong performance of `vrp_cubic` ($R^2 = 4.21\%$) in our results directly echoes the cubic drift specification that is central to their nonlinear model.

4 Feunou, Lopez Aliouchkin, Tédongap, and Xi (2019): Variance Premium, Downside Risk, and Expected Returns

4.1 VRP Decomposition

Feunou et al. (2019) decompose the individual stock VRP into a “bad” (downside) and “good” (upside) component:

$$VRP_{i,t} = VRP_{i,t}^{bad} - VRP_{i,t}^{good} \quad (3)$$

where VRP^{bad} captures the cost of insuring against downside variance (variance from negative returns), and VRP^{good} reflects the discount investors accept for exposure to upside variance. This decomposition is itself a form of *nonlinear transformation*: it applies an asymmetric truncation (indicator function $\mathbf{1}_{r < 0}$) to the return series before computing variance.

The risk-neutral expectations of semi-variances are constructed from option prices following Kilic and Shaliastovich (forthcoming), while realized semi-variances are computed from high-frequency returns.

4.2 Fama–MacBeth Regression Design

This paper provides the most complete template for VRP-based Fama–MacBeth regressions. Their Table 8 presents seven nested model specifications:

Table 1: Fama–MacBeth Model Specifications in Feunou et al. (2019)

Model	Regressors	Key Result
I	Total VRP only	Insignificant at 95%
II	VRP^{bad} only	Significant (positive)
III	$VRP^{bad} + VRP^{good}$	VRP^{bad} significant; VRP^{good} insignificant
IV	$VRP^{bad} + \text{Market } \beta$	VRP^{bad} remains significant
V	$VRP^{bad} + \text{Market } \beta + \text{Size} + \text{B/M}$	VRP^{bad} remains significant
VI	$VRP^{bad} + \text{FF5 betas}$	VRP^{bad} remains significant
VII	$VRP^{bad} + \text{FF5 betas} + \text{characteristics}$	VRP^{bad} remains significant

4.3 Key Findings

1. **Raw VRP is weak in the cross-section:** Model I shows that total VRP alone is *not* statistically significant at the 95% confidence level. This is a crucial finding that resonates with our result that raw `vrp` has modest predictive power ($R^2 = 2.15\%$).
2. **Bad VRP is the driver:** The bad (downside) component produces an annualized risk-adjusted alpha of 18% in portfolio sorts and remains significant through all Fama–MacBeth specifications. A one-standard-deviation increase in bad VRP is associated with a 6.7% to 12.7% increase in annualized expected excess returns.

3. **Independence from standard factors:** The pricing power of bad VRP survives controls for market beta, size, book-to-market, momentum, illiquidity, idiosyncratic volatility, and individual skewness—precisely the set of controls relevant for our planned four-factor regression.

4.4 Relevance to Our Study

The finding that *total* VRP is insignificant while its *transformed* components are highly significant provides a direct parallel to our result. In our case, the transformation is statistical (Z-score standardization) rather than directional (bad/good decomposition), but the mechanism is analogous: raw VRP conflates signals that cancel out or are obscured by heteroskedasticity, while the transformation isolates the informative component.

For our Fama–MacBeth regression, we can adopt Feunou et al.’s progressive specification strategy:

- Start with the VRP factor alone (raw and Z-score versions)
- Sequentially add market beta, size, and book-to-market controls
- Compare the significance and magnitude of the VRP risk price across specifications

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